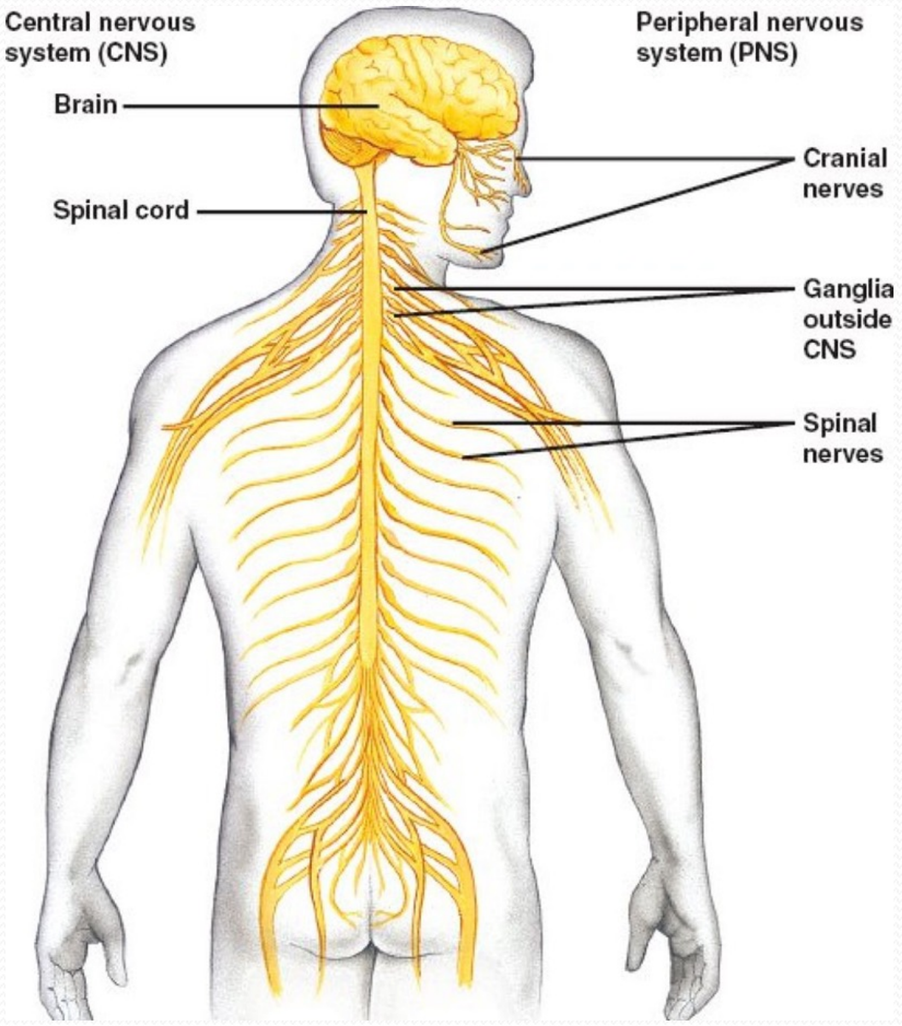
- The nervous system

- The nervous system together with the endocrine system controls and integrates the activities of the different parts of the body.

- The nervous system is divided into two main parts, the central nervous system, which consists of brain and spinal cord, and the peripheral nervous system which consists of 12 pairs of cranial nerves and 31 pairs of spinal nerves and their associated ganglia.



- Functionally the nervous system can be further divided into the somatic nervous system which controls voluntary activities and autonomic nervous system which controls involuntary activities.

# Central nervous system

- It consists of the brain and spinal cord.
- The brain and spinal cord are composed of gray matter and white matter.
- The nerve cell bodies constitute the gray matter.
- The axons form the white matter.
- Nucleus is composed of a group of cell bodies with same function inside the central nervous system.
- A bundle of axons inside the CNS is named as tract.

- The brain is inside the cranial cavity by which it is well protected.
- The spinal cord lies within the vertebral canal by which it is well protected.
- Brain and spinal cord have three coverings ,from outer to inner they are dura matter, arachnoid matter and pia matter. (the meninges)

# Peripheral nervous system

- Is composed of
- A.) 12 pairs of cranial nerves. (they arise from the brain)
- B.) 31 pairs of spinal nerves. (they arise from the spinal cord)
- C. )Ganglia( which are composed of collections of nerve cell bodies outside the CNS)

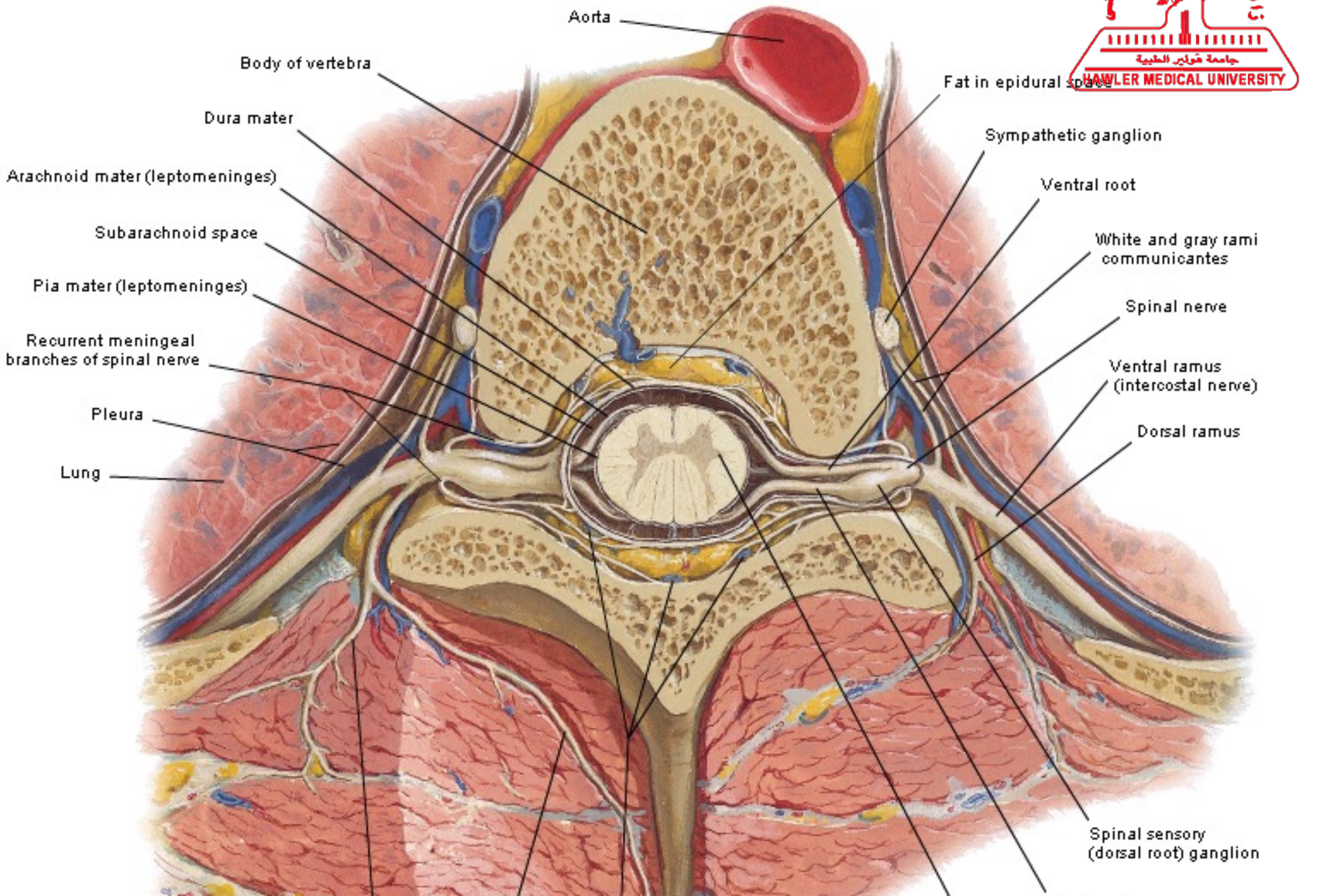
# Structure of the peripheral nerves

- A peripheral nerve (either cranial or spinal) is composed of bundles of nerve fibers (axons) whose cell bodies lie within the gray matter of CNS or in a ganglion.
- In any nerve, each nerve fiber (axon) is surrounded by series of schwann cells which either myelinate it (provide it with a myelin sheath) or do not.( in unmyelinated nerve fibers.
- myelin sheath is derived from the plasma membrane of the shcwann cells.

- The spinal cord is composed of 31 spinal segments which are 8 cervical 12 thoracic, 5 lumbar, 5 sacral and one coccygeal spinal segments.
- Each segment gives rise to a pair of spinal nerve. So there are 8 cervical, 12 thoracic, 5 lumbar, 5 sacral and one coccygeal spinal nerve on each side.
- In cross section the spinal cord have a peripherally located white matter and at the center there is an H shaped gray matter.

# Cross Section through Thoracic Vertebra

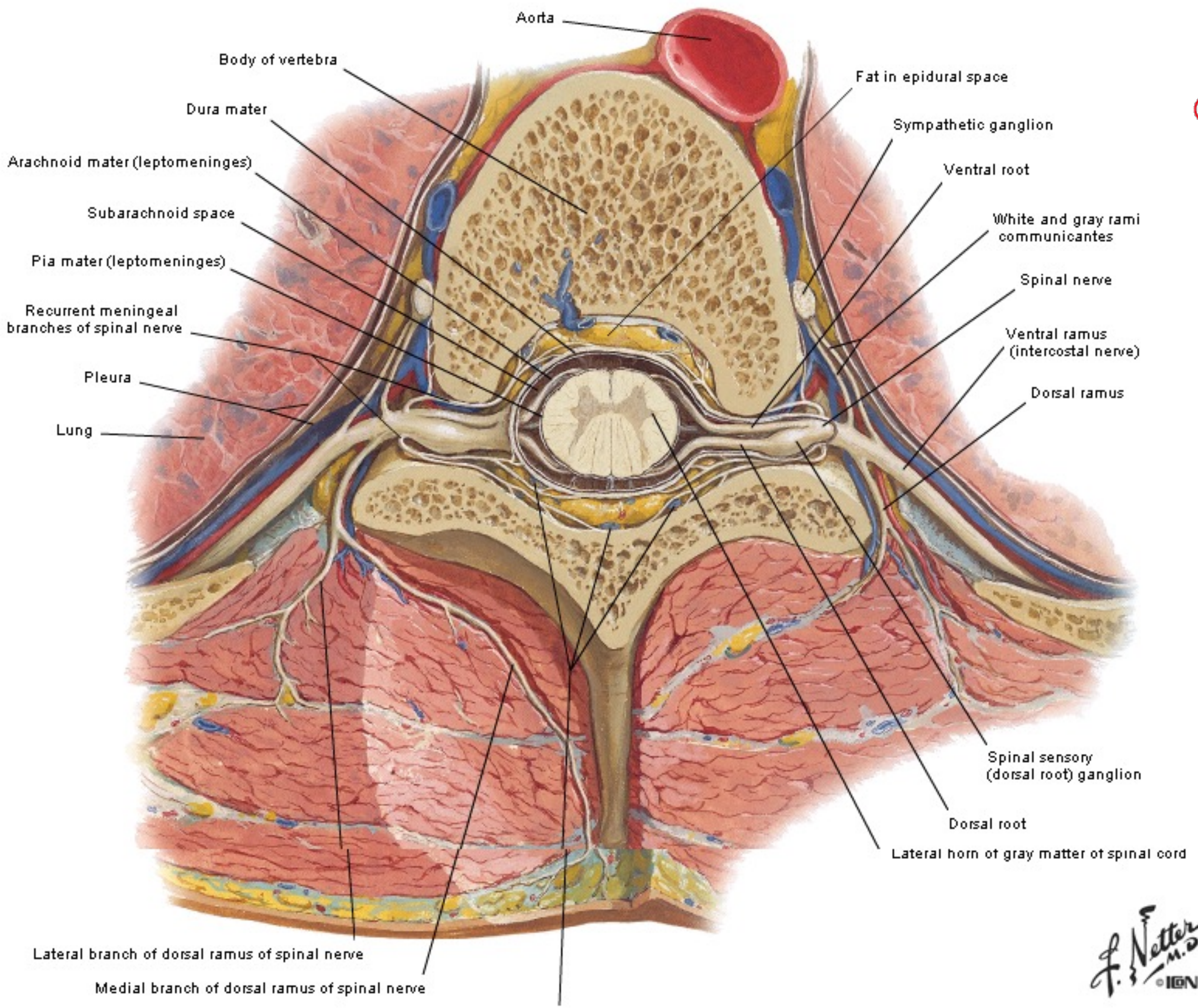




- Thus there are two anterior horns (composed of motor nuclei )and two posterior horns (composed of sensory nuclei)
- In addition to these, there are lateral horns in the spinal segments T1 to L2. ( nucleus of sympathetic preganglionic neurons)
- Also there are lateral horns in S2 S3 and S4 segments (nucleus of parasympathetic preganglionic neurons)

- Each spinal nerve arise by two roots, anterior root which is the motor (efferent) root, and posterior root which is sensory root. (afferent)
- In the dorsal root there is dorsal root ganglion which contains sensory neurons.
- The two roots unite to form the trunk of the spinal nerve which immediately divide into ventral ramus and dorsal ramus of the spinal nerve.
- The ventral ramus supplies anterior and lateral parts of the trunk and the limbs.
- The dorsal ramus supplies the posterior part of the trunk.

## Cross Section through Thoracic Vertebra



- The brain

# BRAIN

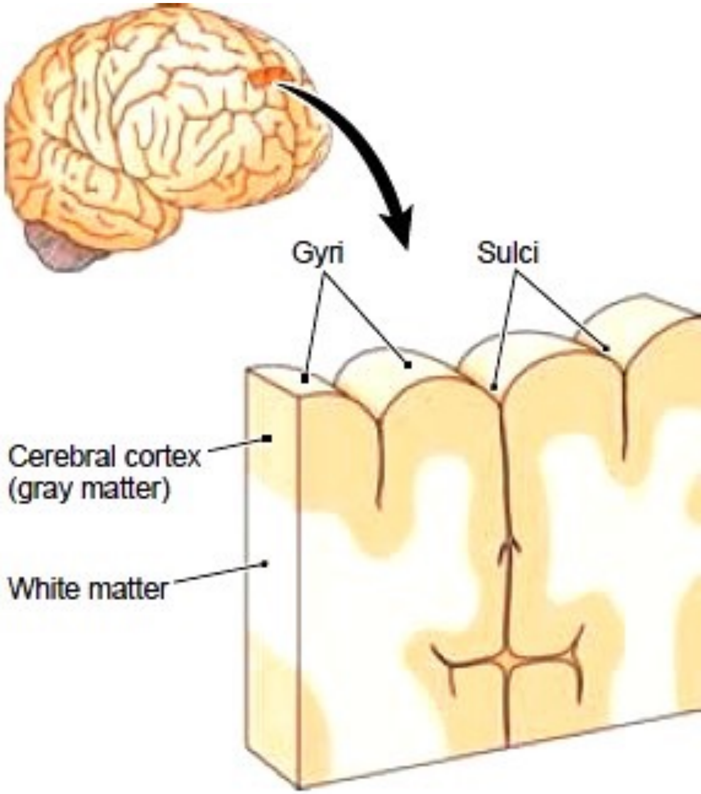
Brain

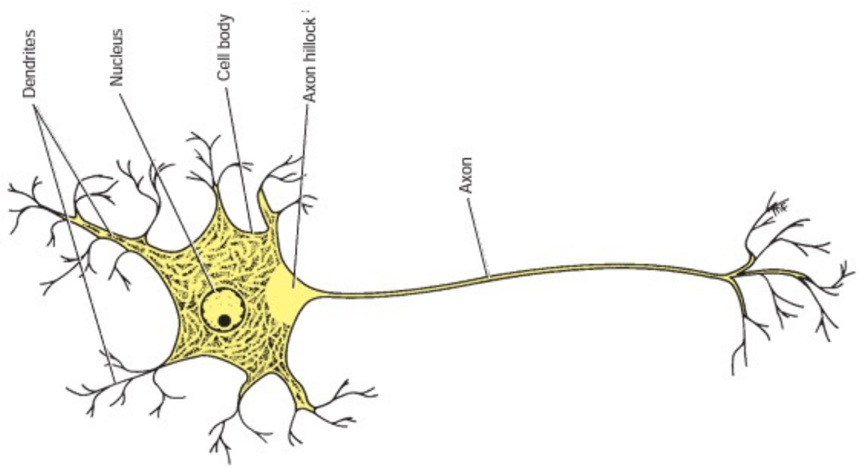
Cerebrum

Cerebellum

Brainstem

The cerebral cortex is composed of ***bodies*** of 10 billions of neurons (gray matter), while situated below the cortex is the white matter, which is formed by the ***axons*** of 10 billions of neurons(white matter)



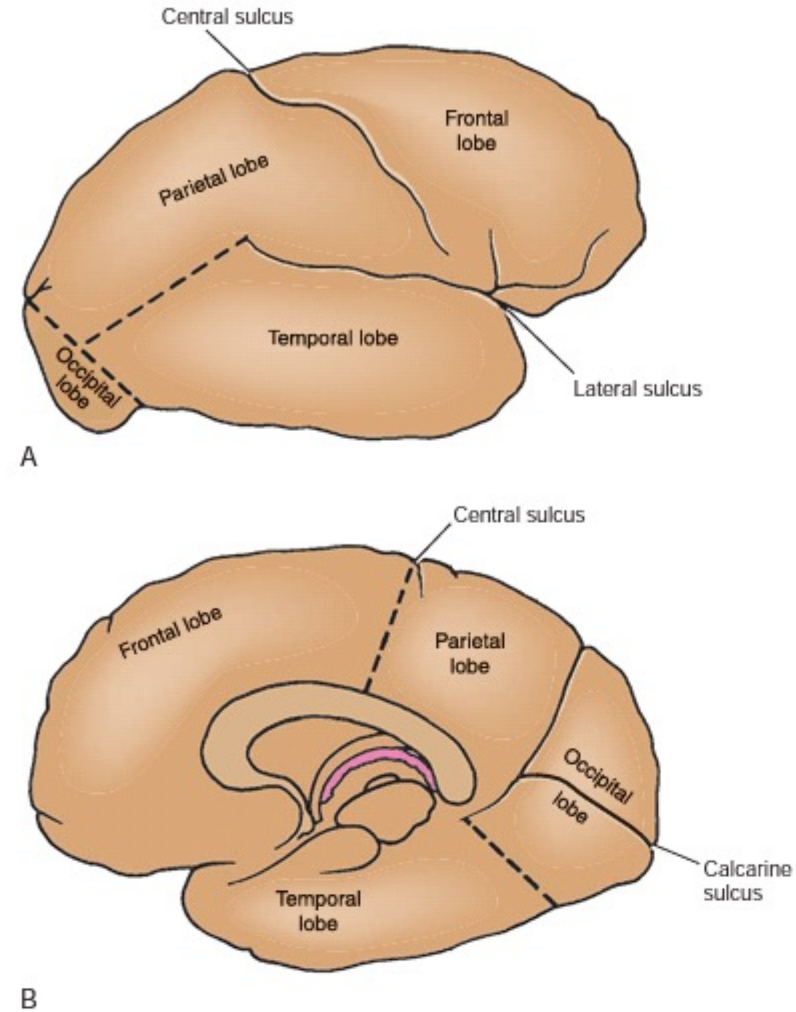
# Cerebrum

The cerebral cortex, the outermost portion of the cerebral hemisphere, is a gray matter structure characterized by many convolutions that greatly increase its surface area.

The bulges or eminences referred to as gyri, and the spaces separating the gyri are called sulci.

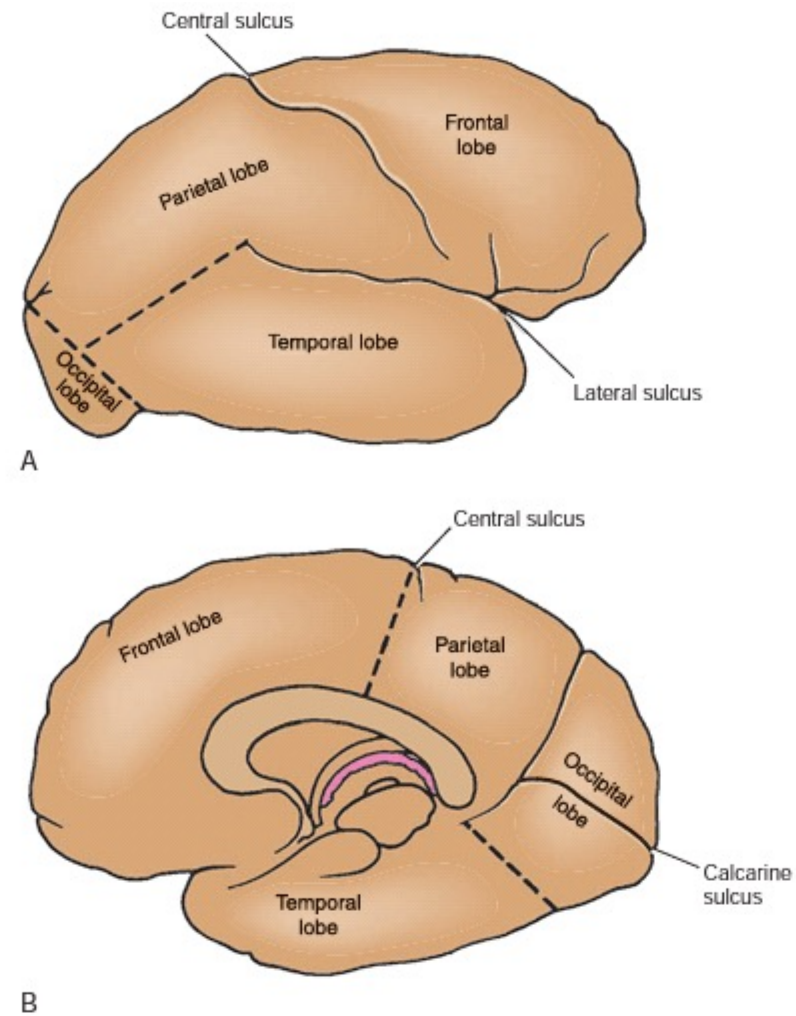
# Important sulci:

1. Central sulcus
2. Lateral sulcus



- Lobes:

1. Frontal
2. Parietal:
3. Temporal:
4. Occipital:

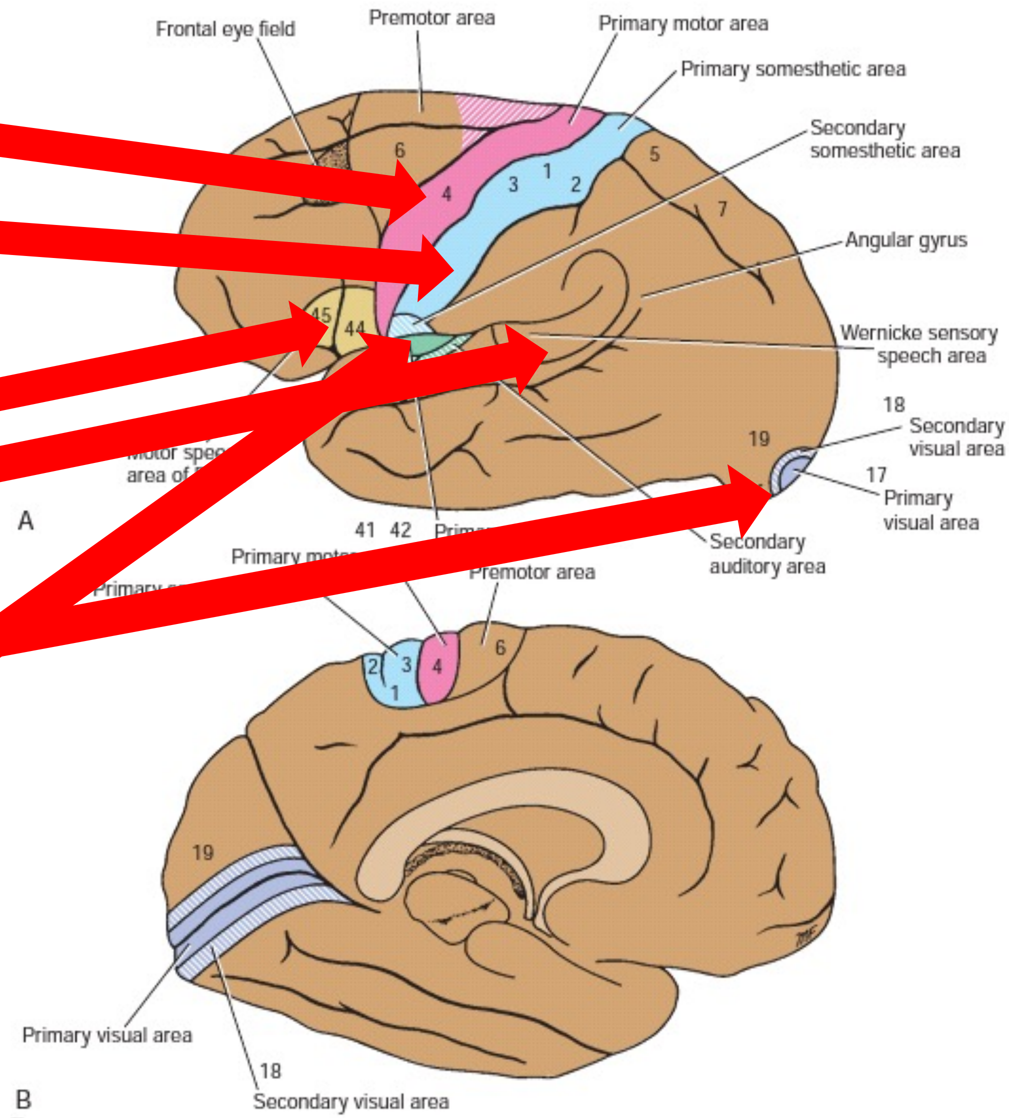


# major gyri and important cortical areas:

1. precentral gyrus (is motor area)
2. Post central gyrus (sensory area)
3. Near the calcarine sulcus(visual area)
4. Brocas area (motor speech area)
5. Wernicke area (sensory speech area)

# Important cortical areas

- Motor area
- Sensory area
- Speech areas
  - Broca
  - Wernickie
- Vision area
- Hearing area



# Lesions affecting specific cortical areas

- **Motor area**: contralateral(opposite side) weakness of the upper and lower limbs
- **Sensory area**: loss of sensation on the contralateral upper and lower limbs
- **Broca's area**: aphasia(understands speech but can't speak)
- **Wernickie's area**: aphasia(can speak but not understand speech)
- **Vision area**: blindness

# Cerebral Dominance

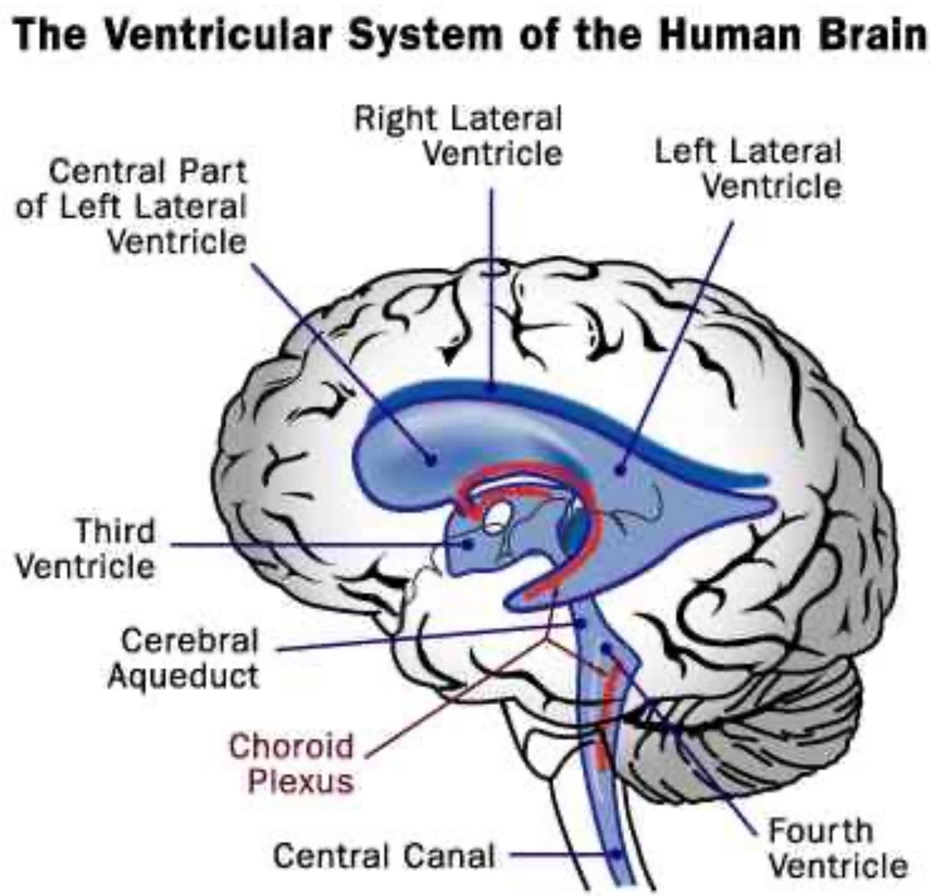
- More than 90% of the adult population is right-handed and, therefore, is left hemisphere dominant.

# Anatomy of Brain: deep structures

- **Thalamus:**
- situated on lateral sides of third ventricle
- Function: sensory relay station
- **Hypothalamus:**
- situated below thalamus, extends from optic chiasm to mammillary bodies
- Function: control of body temperature, eating, drinking, autonomic functions….etc

# Ventricular system and Cerebrospinal Fluid

- **Ventricles of the Brain:**
- 2 Lateral Ventricles
- Third Ventricle
- Fourth ventricle



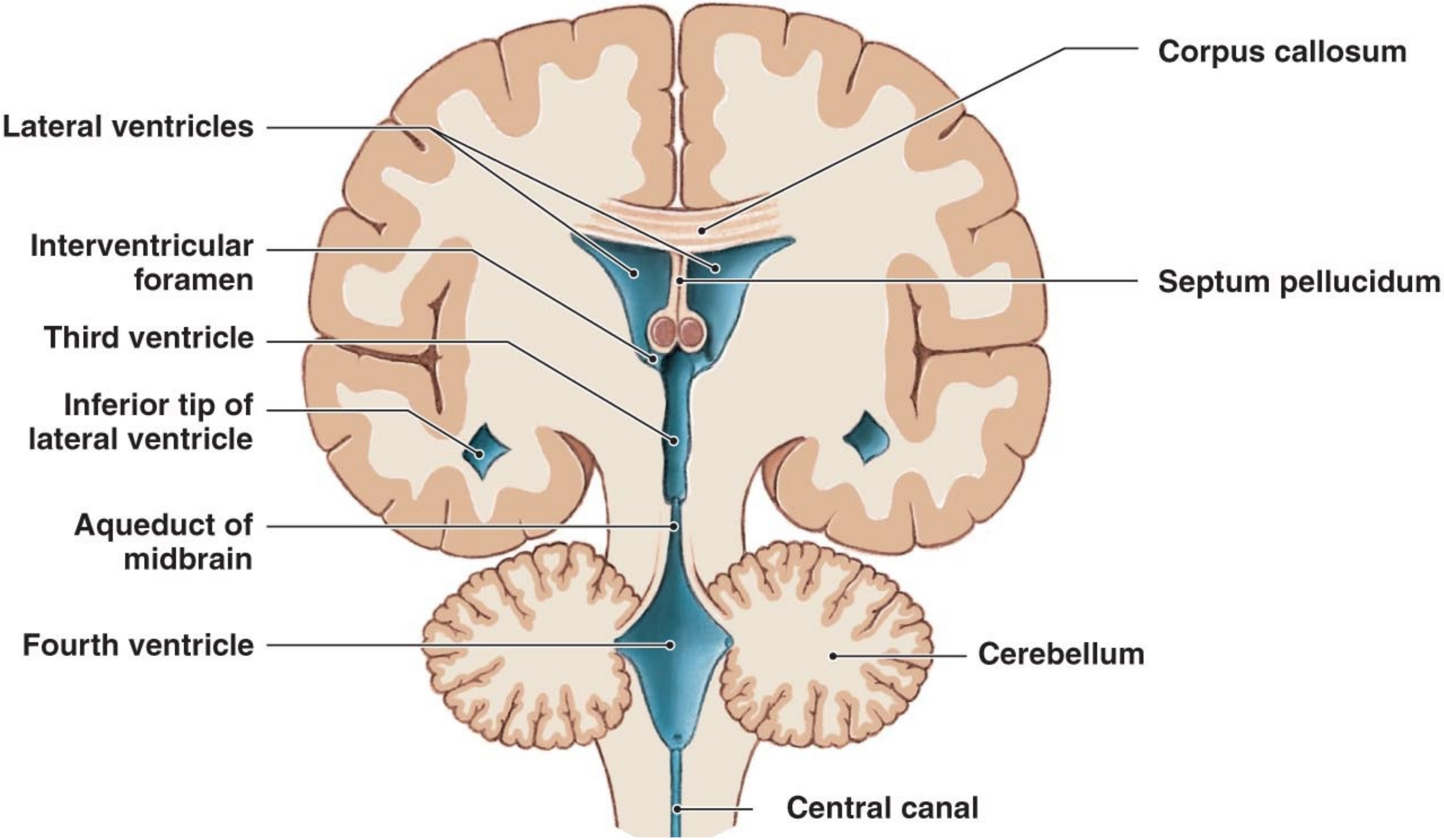
# Cerebrospinal Fluid

- The Cerebrospinal Fluid(CSF) is a clear, colorless liquid that fills the ventricles of the brain and bathes the external surface of the brain and spinal cord.
- It is formed from the choroid plexus within the ventricles

# Circulation of CSF

- From lateral ventricles to third ventricle through foramen of Monro.
- From third ventricle to fourth ventricle through aqueduct of Sylvius
- From fourth ventricles to subarachnoid space through midline formaen of Magendie and lateral foramen of Luschka

Two views of the ventricles, which are filled with cerebrospinal fluid

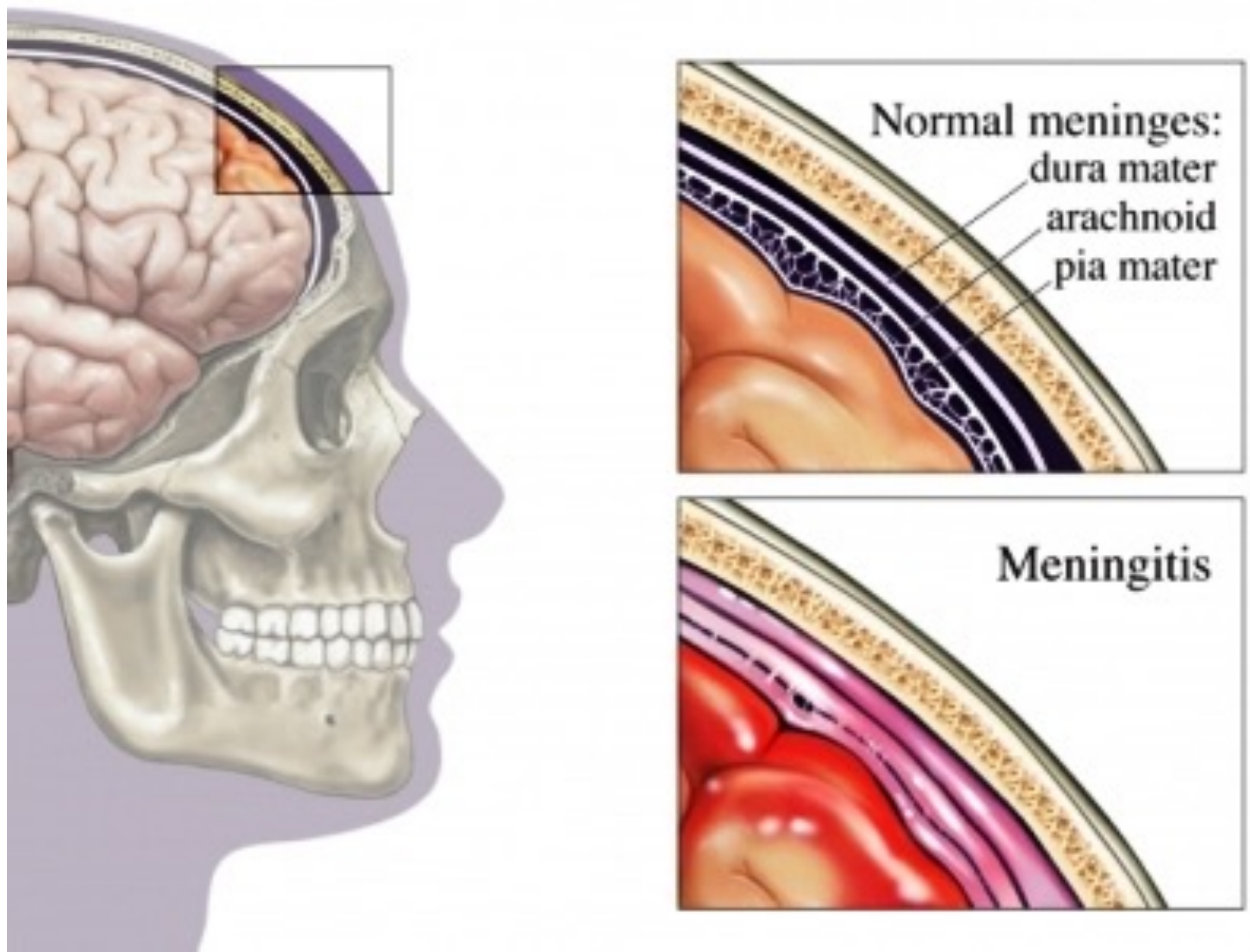


# Functions of CSF:

1. Cushions and protects the central nervous system(CNS) from trauma
2. Provides mechanical support for the brain
3. Assists in the regulation of the contents of the skull
4. Nourishes the CNS
5. Removes metabolites from CNS

# Meningeal Linings of the Brain:

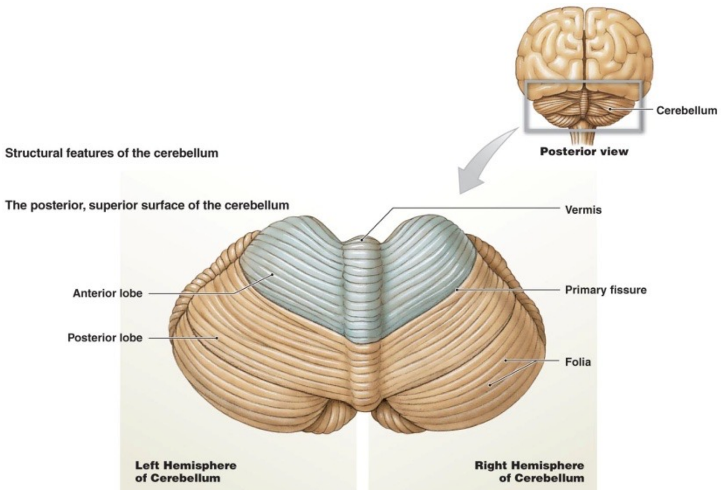
1. Dura mater
2. Arachnoid
3. Pia mater



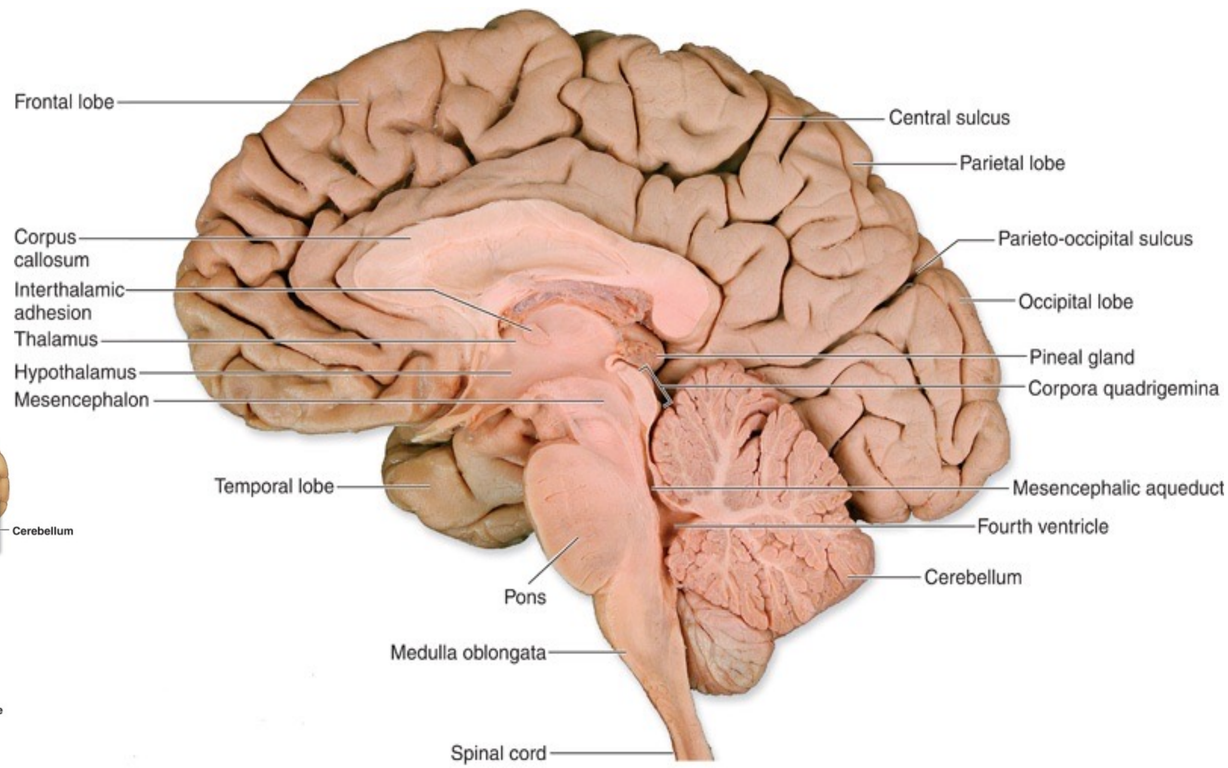
# Diseases related to meninges

- **Meningitis**: it is inflammation of the meninges, usually caused by bacteria.
- Clinical features: Headache, fever and neck stiffness.
- **Epidural Hematoma**: is collection of blood between skull and dura mate. Caused by trauma.
- **Subdural hematoma**: is collection of blood between dura mater and arachnoid, may be caused by trauma or occurs spontaneously.

# Cerebellum:


Structural features of the cerebellum
Cerebellum
Posterior view
The posterior, superior surface of the cerebellum
Vermis
Anterior lobe
Primary fissure
Posterior lobe
Folia
Left Hemisphere of Cerebellum
Right Hemisphere of Cerebellum




Frontal lobe
Central sulcus
Parietal lobe
Corpus callosum
Parieto-occipital sulcus
Interthalamic adhesion
Occipital lobe
Thalamus
Pineal gland
Hypothalamus
Corpora quadrigemina
Mesencephalon
Temporal lobe
Mesencephalic aqueduct
Fourth ventricle
Cerebellum
Pons
Medulla oblongata
Spinal cord

# Cerebellum:

- The cerebellum located in the posterior cranial fossa, behind the brain stem.
- It is composed of

1. Cerebellar hemispheres
2. Vermis

Function: coordination of movements of the body

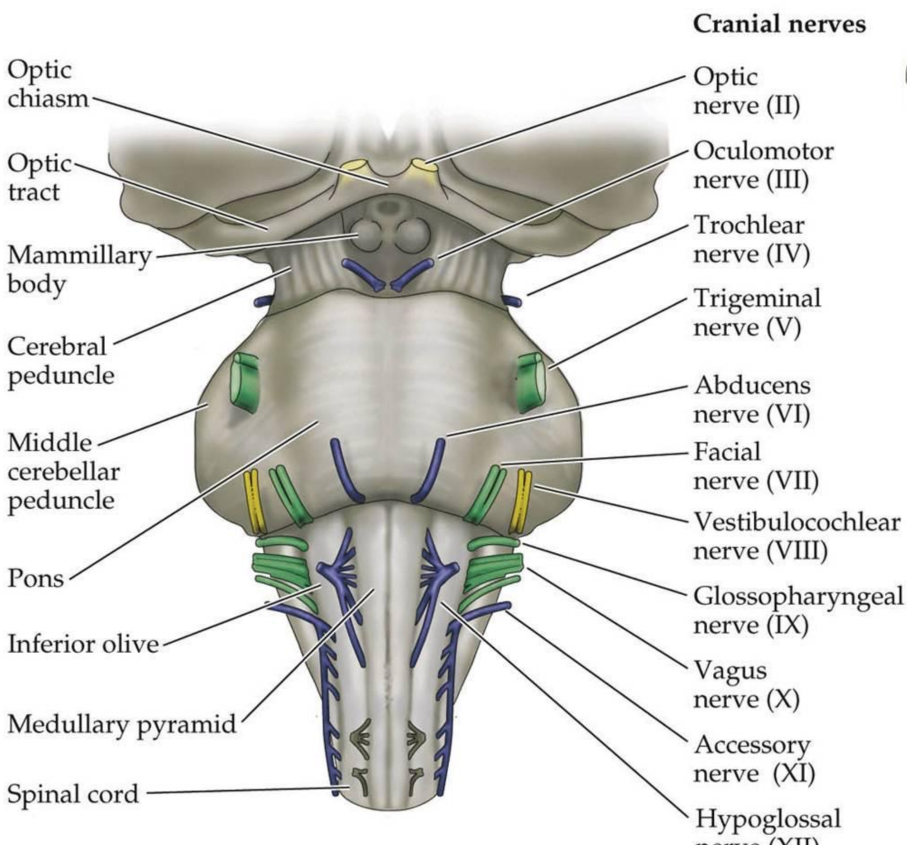
# Brainstem:

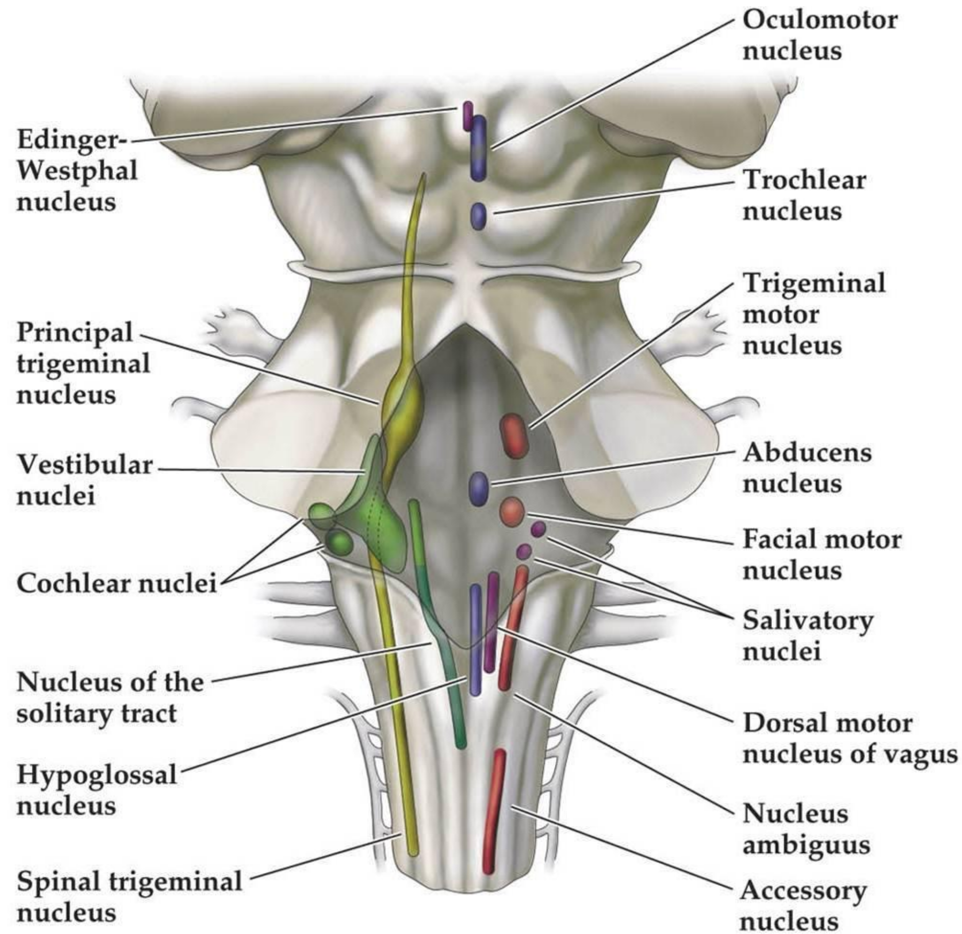
1. **Midbrain**
2. **Pons**
3. **Medulla**

The brain stem contains:

- Nuclei of cranial nerves
- Descending tracts from brain
- Ascending tracts to brain
- Reticular formation

# Brainstem: cranial nerves and nuclei



# Brainstem

The 12 cranial nerves are:

1. Olphactory nerve.
2. Optic nerve
3. Occulomotor
4. Trochlear.
5. Trigeminal
6. Abducent.
7. Facial nerve.

- 8. vestibulocochlear
- 9. glossopharyngeal
- 10. vagus.
- 11. accessory
- 12. hypoglossal nerve.

# Blood supply to the brain

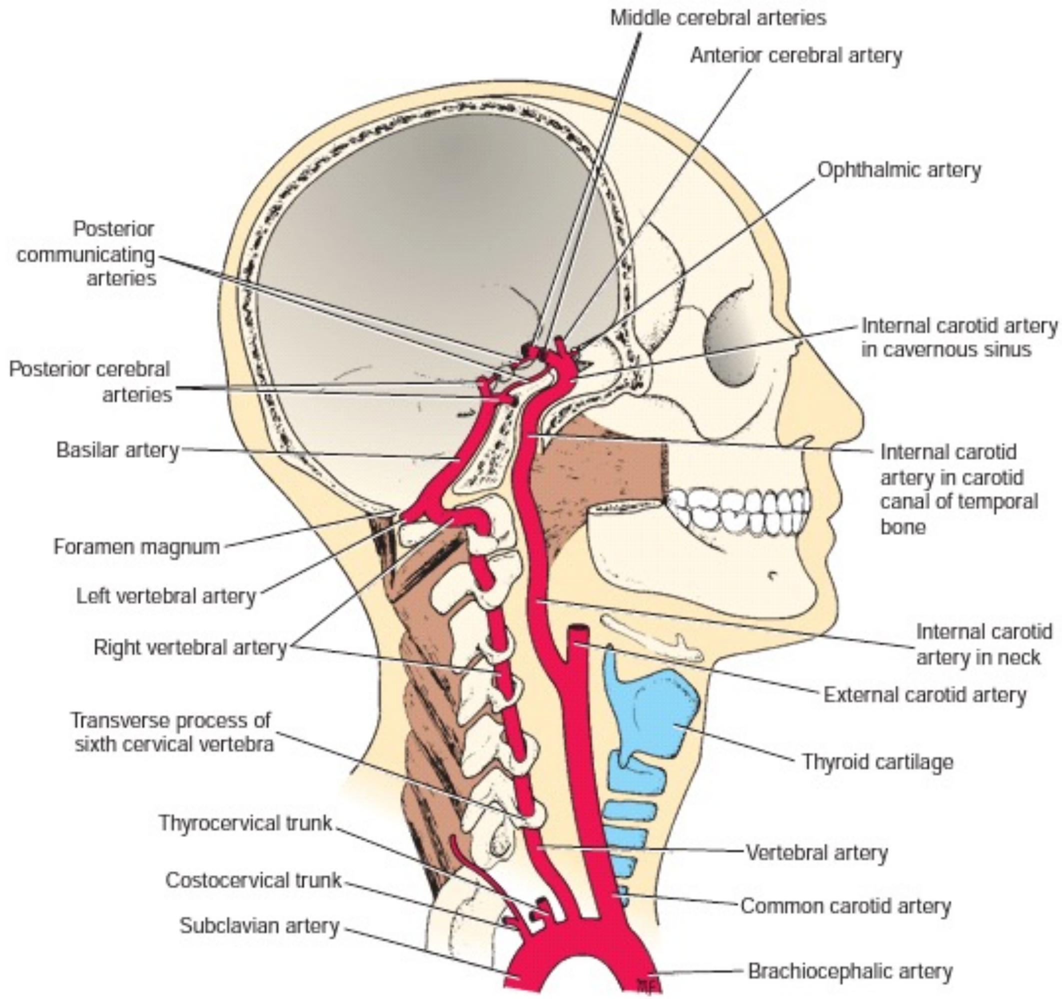
1. **Internal carotid artery:** divides to:

- Anterior Cerebral Artery(ACA)
- Middle Cerebral Artery(MCA)

2. **Vertebral artery:** they join to form Basilar Artery. Then the Basilar artery divides to two Posterior Cerebral arteries(PCA).

The internal carotid arteries and posterior cerebral arteries joined by Posterior communicating artery to form the Circle of Willis.

# Blood supply to the brain



# Venous Drainage of the Brain

- Superficial cerebral veins
- Deep cerebral veins
- Venous sinuses:

1. Superior sagittal sinus
2. Inferior sagittal sinus
3. Straight sinus
4. Transverse sinus
5. Sigmoid sinus

They drain into internal jugular vein

# Venous Drainage of the Brain